

COMPARISON OF KOLMOGOROV-ARNOLD NEURAL NETWORKS WITH NON-UNIFORM RATIONAL B-SPLINES TRAINABLE ACTIVATION FUNCTIONS WITH PHYSICS INFORMED NEURAL NETWORKS AND ISOGEOMETRIC FINITE ELEMENT METHOD SOLVERS FOR ADVECTION-DOMINATED DIFFUSION PROBLEMS

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ABSTRACT

We present a comparison of Kolmogorov-Arnold Neural Networks [1] with Non-Uniform Rational B-splines trainable activation functions (IGA-KANN) with Physics-Informed Neural Networks (PINN) and higher-order and continuity Isogeometric Finite Element Method (IGA-FEM). We focus on the two-dimensional Poisson problem with manufactured solution, the first one with $\sin(2x\pi)\sin(2y\pi)$ exact solution, the second one with $\sin(2x\pi)\sin(2y\pi)\exp((x-2y)\pi)$, as well as the two-dimensional advection-dominated diffusion problem including the Eriksson-Johnson model problem.

We first solve all these problems with standard PINN method, and IGA-KAN method, comparing the convergence of the numerical error employing the known exact solution resulting from the manufactured formulation. Next, we employ the robust loss function [2] that multiplies the residual loss by the inverse of carefully constructed Gram matrix. This robust loss approximates the Riesz representative of the residual and it is related with the true error. We run the training for the PINN and IGA-KANN methods with residual loss and robust loss functions, and compare the convergence and accuracy. For IGA-FEM solvers we employ Streamline-Upwind Petrov-Galerkin (SUPG) stabilization method and Isogeometric Residual Minimization methods (IGA-RM) [3].

We compare the training and solutions of PINN, Robust PINN, IGA-KANN and Robust IGA-KANN methods with higher-order IGA-FEM methods. We consider a case with uniform FEM and uniform distribution of points for PINN. We also consider adaptive IGA-FEM, refined towards edge singularity, and non-uniform distribution of points for PINN, as well as non-uniform distribution of points for IGA-KANN.

We conclude that IGA-KANN provides better convergence and numerical accuracy, and it is an advancement of the scientific computing with neural network.

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